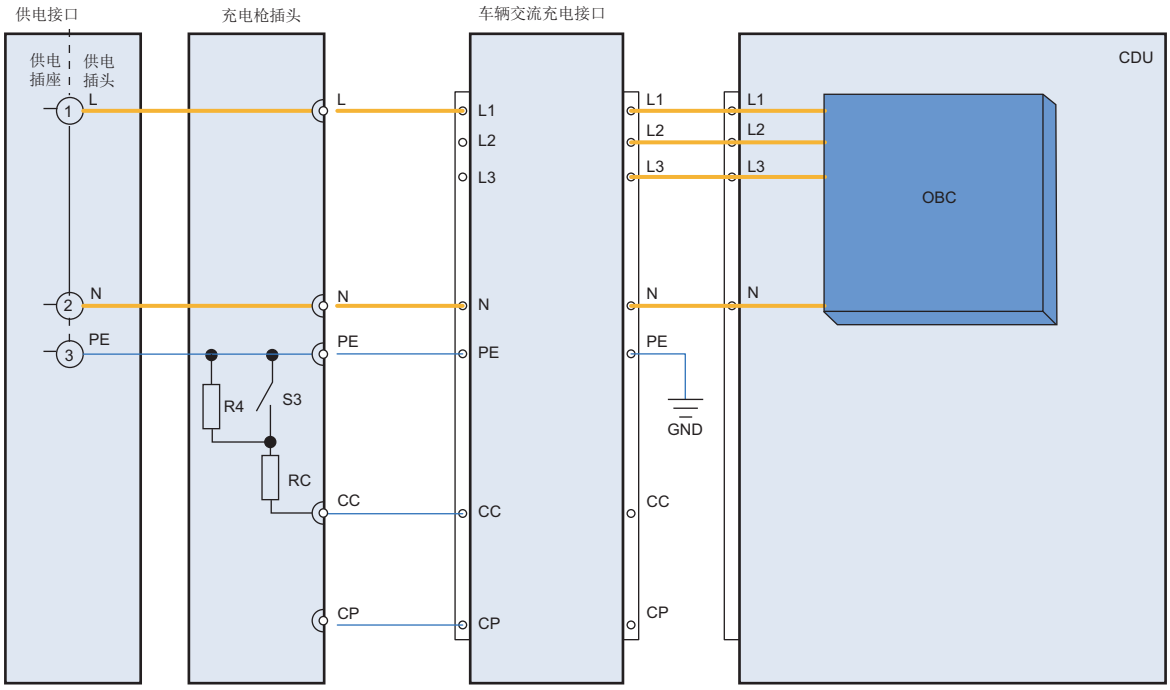
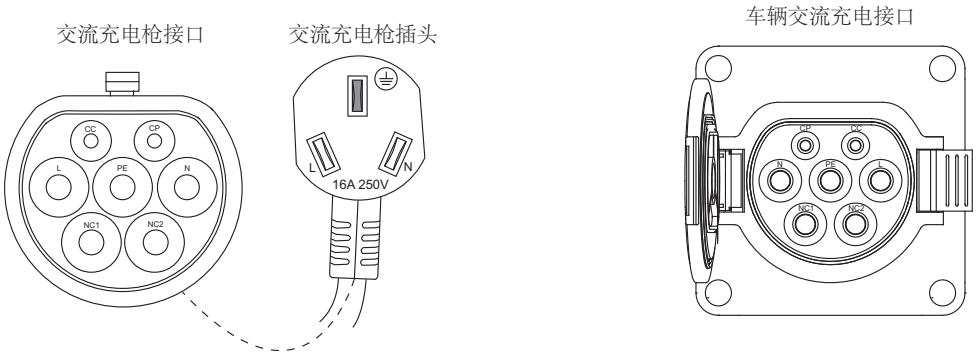
P1F20-00、 P1F21-00、 P1F22-00

Fault diagnosis code
P1F20-00：Inverter AC undervoltage
P1F21-00：Inverter AC overvoltage
P1F22-00：Inverter AC overcurrent
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• Circuit,terminal and adapting piece fault • OBC battery fault • The inverter AC module in the CDU is faulty
Conditions for fault code operation
Turn the ignition switch to ON position.
Conditions for fault code setting
P1F20-00
P1F21-00
P1F22-00
To set the effect of a fault code condition
Description of circuit diagram

Circuit diagram





EV904010

Troubleshooting steps		
Step 1	General inspection.	
		1 Check the fuse for damage. 2 Check battery capacity. 3 Check the harness plug of CDU for damage, poor contact, aging and looseness. 4 Check whether the result is normal or not?
		NO → Repair or replace the faulted parts.
YES ↓		
Step 2	Check fault codes.	
		1 Connect the fault diagnosis instrument to the diagnosis interface. 2 Turn the ignition switch to ON position. 3 Diagnose the GAW module with diagnostic apparatus. 4 Is there any other fault code except for P1F20-00/P1F21-00/P1F22-00?
		YES → Refer to:DTC Summary list(CDU).
NO ↓		
Step 3	Check the AC power supply.	
		1 Make protective preparations according to the safety operation specifications of the high voltage voltage in the maintenance manual. 2 Check the input voltage of AC power supply. 3 Check whether the result is normal or not?
		NO → Replace the AC power in good condition.
YES ↓		
Step 4	Check the AC charging gun.	
		1 Check whether the AC charging status is intact and the function is normal. 2 Check whether the result is normal or not?
		NO → Repair or replace the faulted parts.
YES ↓		

Step 5	Check the cable connecting the cahrging port to the OBC.
	1 Turn the ignition switch to OFF. 2 Disconnect the high voltage wiring harness connector of the vehicle. 3 Check whether the charging port to the OBC connection cable is rotten, damaged, and water enters. 4 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	
Step 6	Check inverter AC circuit.
	1 Turn the ignition switch to OFF. 2 Make protective preparations according to the safety operation specifications of the high voltage voltage in the maintenance manual. 3 Open the CDU end cover. 4 Check inverter AC circuit. 5 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	
Step 7	Check OBC.
	1 Turn the ignition switch to OFF. 2 Make protective preparations according to the safety operation specifications of the high voltage voltage in the maintenance manual. 3 Open the OBC end cover. 4 Check whether the internal circuit of OBC is damaged. 5 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 8	Test whether CDU is norm.
	1 Turn the ignition switch to OFF. 2 Disconnect and check the harness plug EB1004、BT1001、BTN_1001、BTN_1002 of CDU. 3 Check and repair. • Corrosion(install new adapting piece or terminal-cleaning module and pin). • Pin damaged or bent-install new terminal/pin. 4 Fly out of pins-install new pins as needed. 5 Reconnect the CDU plug and all other previously disconnected plug. Make sure the pin is properly positioned and fixed. 6 Run the system and determine if the problem persists?
	YES → Change the CDU Module
NO ↓	
At this point the system is functioning normally. The problem may be caused by module connections. Address the root cause of any connector or pin problems.	
Next Step ↓	
Diagnosis end.	